Reproducing Deepfake Detection Using Multimodal CLIP-Based SigLIP-2 Vision Transformers

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GitHub: https://github.com/Salar-khan-bits/Deepfake_Detection

Abstract

Reliable deepfake detection is now a practical requirement because forged media spreads faster than manual verification can keep up. Soundararajan and Xu (2026) addressed this problem with SigLIP-2 Vision Transformers on CiFake, HiDF, and SID_Set. We reproduce their pipeline under a single Google Colab T4 constraint by replacing ViT-L/16 with ViT-B-16-SigLIP-384 and reducing memory-heavy settings. Our setup reaches 91.93% accuracy on CiFake (10K), 79.60% on HiDF binary evaluation, and 85.94% on SID_Set binary classification, while few-shot transfer from HiDF features gives 84.96% on Deepfake-vs-Real-60K. In Assignment 3, scaling CiFake to 140K raises accuracy to 94.40%, but cross-domain tests remain near chance, which indicates that our heads still learn dataset-specific cues rather than broadly transferable forgery evidence.

1 Introduction

Synthetic media quality has improved quickly, and forged visual content now appears in ordinary news and social feeds. That shift raises the risk of misinformation, targeted harassment, and fabricated evidence. Human inspection alone is no longer reliable once manipulations are post-processed and distributed at scale. Automated detection therefore remains a central problem in computer vision.

Earlier detectors relied on CNNs that focused on local artifacts from specific generators. They often worked on matched test data and then dropped sharply on unseen methods or different compression settings. The core issue is generalization across forgery families, not only accuracy on one benchmark split. A practical detector must transfer from one manipulation style to another without retraining from scratch.

Soundararajan and Xu [1] propose a SigLIP-2 ViT framework that addresses this transfer problem through large-scale contrastive pretraining. Their model freezes most of ViT-L/16 and trains a lightweight head, yet still reports high results on CiFake, HiDF, and SID_Set. They also report effective few-shot adaptation to new datasets through class prototypes. The key claim is that pretrained SigLIP-2 features encode authenticity cues that remain useful across tasks.

Under our assignment constraints, we test whether that claim still holds on limited hardware. We train on a Google Colab T4 GPU with 15GB VRAM, so we replace ViT-L/16 with ViT-B-16-SigLIP-384 and lower memory-intensive settings such as batch size. We also use reduced data where full-scale training is impractical. These decisions let us run the full pipeline while preserving the paper's main learning objective.

Beyond reproduction, we add three targeted extensions. We first scale CiFake from 10K to 140K images to measure how much data can offset a smaller backbone. We then evaluate cross-domain transfer by testing trained models on a stylistically different forgery dataset. We finally adapt SID_Set to binary classification to quantify what is lost when we remove the segmentation branch.

Section 2 connects prior generation and detection work to the gap we study. Section 3 describes the original model and our constrained variant. Section 4 states datasets, preprocessing, and training choices. Section 5 reports reproduction and extension results, Section 6 interprets the main patterns, and Section 7 closes with implications and next steps.

2 Background and Related Work

Deepfake generation moved from hobbyist autoencoders to high-fidelity adversarial models within a short period. StyleGAN [3] and StarGAN [4] raised photorealism, while reenactment systems such as Face2Face [2] made identity and expression transfer practical in near real time. As generation quality improved, artifact-based detection became less dependable. The field then shifted toward representations that model global consistency rather than

only local defects.

CNN-based detectors were the first strong baseline family. They captured blending seams, color inconsistencies, and compression traces, and they reached solid in-domain accuracy on early benchmarks. Performance usually declined when data sources or forgery pipelines changed, which exposed a narrow inductive bias toward training artifacts. That limitation motivated architectures with broader contextual modeling.

Vision Transformers became a natural next step because self-attention captures long-range dependencies across the full frame. Work such as TALL [10] and related transformer variants reported better behavior on harder detection settings, including video-oriented tasks. These gains suggested that deepfake cues often span spatial regions that CNN kernels do not connect effectively. Transformer backbones therefore became a common foundation for newer detectors.

CLIP-style pretraining introduced another progression in this line of work. FatFormer [9] and DeCLIP [8] indicate that image-text contrastive training yields features that transfer to authenticity prediction with limited task-specific tuning. The likely reason is that contrastive objectives enforce semantically coherent embedding geometry across diverse web data. Fake images then appear as structured deviations from that geometry, even when artifact patterns differ by generator.

SigLIP-2 extends this idea with a sigmoid-based training objective on WebLI and scales effectively to very large data regimes. In the Soundararajan and Xu setup [1], ViT-L/16 at 384×384 yields rich token features that support both classification and adaptation. On the benchmark side, HiDF [6], SID_Set [7], and CiFake [5] cover different forgery types and evaluation protocols, which makes them a strong combined stress test. The open question is whether similar behavior survives when compute forces a smaller backbone and stricter training limits.

Our study fills that reproduction gap. We keep the core SigLIP-2 strategy, apply it under constrained hardware, and quantify where performance is preserved or lost across in-domain, few-shot, and cross-domain settings.

3 Proposed Model

3.1 Original Paper Architecture

Soundararajan and Xu [1] build their detector on SigLIP-2 ViT-L/16 pretrained with OpenCLIP on WebLI. The model receives 384×384 RGB input, forms a 24×24 grid of 16×16 patches, and produces 1024-dimensional features. They freeze blocks 1–21 and fine-tune only the final transformer blocks with the task head. This design keeps adaptation capacity while controlling optimization cost.

The classification head uses LayerNorm, a 1024-to-512 projection, GELU, dropout at 0.2, and a final binary logit layer. Before classification, feature vectors receive Gaussian noise with $\sigma = 0.02$ and L2 normalization. That regularization discourages dependence on raw embedding magnitude. The head therefore learns boundaries from direction-aware feature structure instead of scale artifacts.

For CiFake and HiDF the training process uses BCEWithLogitsLoss. For SID_Set, they added a SegFormer-style decoder. This helped in three-class prediction and finding tampered regions. The overall goal combines two things: BCE and Dice terms.These help to check the segmentation of the images. In this setup tasks link class prediction to spatial proof.

The few-shot part takes out the trained head. Makes class examples, from frozen encoder results. Given asupport set of k images per class, prototype $\mathbf{c}_j$ for class j is computed as:

$$\mathbf{c}_j = \frac{1}{k} \sum_{i=1}^{k} \frac{f(\mathbf{x}_i)}{\|f(\mathbf{x}_i)\|_2} \tag{1}$$

This equation calculates the mean normalized embedding for class j.

Query images are then assigned probabilities from distance to each prototype with temperature scaling:

$$p(y = j \mid \mathbf{x}) = \frac{\exp(-d(\mathbf{x}, \mathbf{c}_j)/\tau)}{\sum_{j'} \exp(-d(\mathbf{x}, \mathbf{c}_{j'})/\tau)} \tag{2}$$

This equation converts prototype distances into class probabilities through a softmax over negative distances.

Training uses AdamW with learning rate 2×10^{-4} and weight decay 0.01. The paper reports 20 epochs on CiFake/HiDF and 24 on SID_Set at batch size 64 using an RTX 5090 (24GB).

3.2 Our Reproduced Architecture

Our hardware limit is a T4 with 15GB VRAM, so ViT-L/16 at the paper settings is not feasible. We use ViT-B-16-SigLIP-384 with about 86M parameters and 768-dimensional features. We freeze the full backbone rather than partially fine-tuning it. That choice is mainly hardware-driven because it lowers memory and stabilizes training on Colab sessions.

In our head, we keep the same overall structure and reduce dimensions to match ViT-B outputs. The stack is LayerNorm, 768-to-256 linear projection, ReLU, dropout 0.1, and a final binary logit layer. We keep Gaussian noise at $\sigma = 0.02$ and L2 normalization since these are central to the original regularization scheme. We also keep AdamW at the same learning rate and weight decay, then add cosine annealing as a training-stability design choice.

Batch size drops from 64 to 32 for memory safety. For initial CiFake reproduction, we train on 10K images instead of the full dataset. These limits make our baseline conservative, which helps isolate what can still be recovered under constrained compute.

3.3 Assignment 3 Extensions

Assignment 3 adds three experiments beyond direct reproduction. We scale CiFake training from 10K to 140K images to test data-volume effects on a frozen-backbone setup. We then run cross-domain evaluation on 140K Real & Fake Faces to measure transfer under distribution shift. For SID_Set, we simplify the original three-class-plus-segmentation task to binary detection because our implementation excludes the decoder branch.

4 Experimental Setup

4.1 Datasets

Table 1 lists all the datasets that we have in our pipeline. The datasets we use for reproduction are CiFake, HiDF and SID_Set. We use the Kaggle datasets for the transfer tests and the adaptation tests.

CiFake has a lot of pictures some are real. Some are made by artificial intelligence. We use a set of 10,000 pictures from CiFake from the full set which has 140,000 pictures.

HiDF has pictures of faces that're fake and these pictures come from different places.

The SID_Set is also used and it tells us more about the pictures that have been changed it has a complicated system, for labeling the original pictures.

Table 1: Summary of datasets used in experiments.

Dataset	Task	Classes	Size
CiFake (10K)	Binary	Real / AI-generated	10,000
CiFake (140K)	Binary	Real / AI-generated	140,000
HiDF	Binary	Real / Deepfake	varies
SID_Set	Binary†	Real / Fake	varies
140K Real & Fake	Cross-domain test	Real / StyleGAN	2,000
AI vs Human	Few-shot target	Human / AI	79,850
Deepfake-vs-Real-60K	Few-shot target	Real / Deepfake	91,234

†Original paper uses 3-class + segmentation; we use binary.

4.2 Preprocessing

We do the thing to all the images we use. We use the SigLIP OpenCLIP transform to make all the images the size, 384×384. We also make sure all the images have the brightness and color. We flip all the images to the side to make them more different. For SID_Set we do things to the images to make them look more like they were shared on social media. We make the colors more extreme add noise and blur or sharpen the images.

4.3 Training Configuration

Table 2 has all the numbers we used. We had to make some choices about how to use the computers memory and time. We use a group of 8 images at a time when we are working with SID_Set. We also stop working on SID_Set after 3 rounds to make sure the computer does not get too busy. For CiFake and HiDF we try to keep everything the same as the paper so we can compare our results.

Table 2: Training hyperparameters: original paper vs. our reproduction.

Setting	Paper (ViT-L)	Ours (ViT-B)
Backbone	ViT-L/16-SigLIP-2	ViT-B-16-SigLIP-384
Feature dim	1024	768
Hidden dim	512	256
Activation	GELU	ReLU
Dropout	0.2	0.1
Frozen blocks	1–21 (of 23)	All (fully frozen)
Batch size	64	32
Learning rate	2×10^{-4}	2×10^{-4}
Weight decay	0.01	0.01
Noise σ	0.02	0.02
L2 norm	Yes	Yes
Scheduler	None	Cosine Annealing
Epochs (CiFake)	20	10
Epochs (HiDF)	20	20
Epochs (SID)	24	3
GPU	RTX 5090 (24GB)	T4 (15GB)

4.4 Few-Shot Setup

When we want to see if our method can work with a few examples we use the HiDF checkpoint. We remove the part that decides what is in the image. We do not change the part that looks at the image. We pick 50 images from each group. Calculate the average image. We then look at images and decide what is in them by how similar they are to the average images. We use the image and the flipped image to make our decision. We always use the starting point, seed 42.

4.5 Hardware and Framework

We do all our experiments on one NVIDIA T4 computer, in Google Colab. We use PyTorch and other tools like open-clip-torch and Albumentations. We use Python 3.10. Turn on AMP when we can.

5 Results

5.1 Reproduced Results — CiFake

Table 3 compares our CiFake baseline with the paper. On the 10K split, we obtain 91.93% accuracy and 0.9721 AUC, versus paper values above 95% accuracy and 0.986 AUC. The 3.07-point accuracy gap likely comes from the smaller backbone and reduced training data. At the same time, the AUC gap is only 0.0139, which indicates similar ranking quality even when the thresholded boundary is weaker.

Table 3: CiFake binary classification: paper vs. our reproduction.

Metric	Paper	Ours (10K)
Accuracy	>95%	91.93%
AUC	0.986	0.9721
F1	∼0.950	0.9196
AP	∼0.972	0.9712
Best Epoch	—	6

Figure 1 tracks convergence. Validation accuracy levels off near epoch 6, which matches the selected checkpoint and suggests early saturation on the 10K subset.

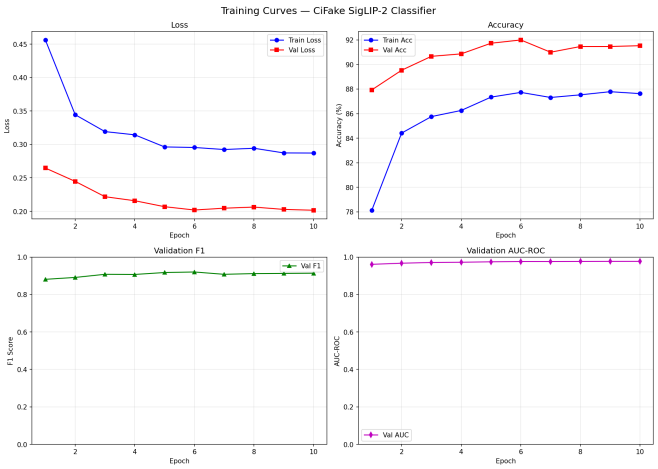

Figure 1: Training curves for CiFake binary classifier (ViT-B-16-SigLIP-384, 10 epochs on T4 GPU).

Figure 2 reports confusion, ROC, and PR views. Errors are fairly balanced, though false negatives for fake samples remain slightly higher, which likely reflects borderline synthetic images that resemble the real class distribution.

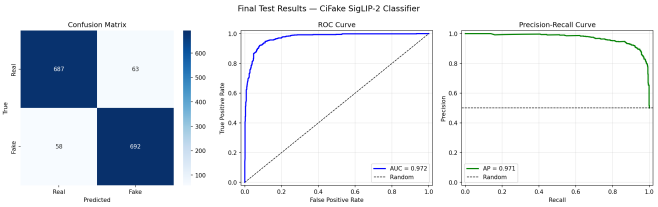

Figure 2: Final test results for CiFake: confusion matrix, ROC curve, and precision-recall curve.

5.2 Reproduced Results — HiDF

HiDF results in Table 4 place our model at 79.60% accuracy and 0.7920 F1. The paper reports about 91% accuracy on this track, so the gap is 11.40 points. One explanation is reduced adaptation capacity, since we freeze all ViT-B blocks while the paper fine-tunes late ViT-L layers. Precision of 0.8077 versus recall of 0.777 also indicates a conservative operating point that misses part of the fake class.

Table 4: HiDF results: paper vs. baselines vs. our reproduction.

Method	Acc	F1	Prec	Recall
Paper (Image)	∼91%	∼0.920	—	—
EB4 Image	—	—	AP=0.722	AUC=0.697
EB4 Video	—	—	AP=0.712	AUC=0.733
Ours (ViT-B)	79.60%	0.7920	0.8077	0.777

5.3 Reproduced Results — SID_Set

Before comparing numbers, we note a protocol mismatch. The paper uses three-class prediction plus segmentation, while we run binary classification without the segmentation decoder. This difference changes both supervision and label structure, so direct metric alignment is limited.

Table 5: SID_Set classification: paper (3-class) vs. our reproduction (binary).

Metric	Paper (3-class)	Ours (binary)
Accuracy	99.09%	85.94%
F1	0.991	0.8645
Precision (Real)	—	0.89
Precision (Fake)	—	0.83
Recall (Real)	—	0.82
Recall (Fake)	—	0.90

Table 6: SID_Set segmentation baselines from the original paper.

Method	Seg F1	Seg IoU	Pixel Acc
Paper (SigLIP-2)	0.822 (Dice)	0.743	98.28%
SIDA-7B	73.9	43.8	—
MVSS-Net	31.6	23.7	—
PSCC-Net	71.3	35.7	—

Under our binary setting, we reach 85.94% accuracy and 0.8645 F1 in 3 epochs at batch size 8. The paper reports 99.09% on the original three-class task, a 13.15-point gap. Collapsing synthetic and tampered into one fake label likely removes class structure that helps separation in the original setup. Recall is 0.90 for fake and 0.82 for real, which indicates higher sensitivity to manipulated samples at the cost of extra real-image alarms.

5.4 Few-Shot Adaptation Results

Table 7 summarizes 50-shot prototype transfer from the HiDF encoder to two unseen datasets. We evaluate each

on a 4,900-image test subset to keep evaluation conditions matched across runs.

Table 7: Few-shot adaptation: paper vs. our reproduction (50-shot, HiDF encoder).

Dataset	Metric	Paper	Ours
AI vs Human	Accuracy	0.8917	0.7716
	F1	0.8917	0.7715
	AUC	0.9553	0.8451
	AP	—	0.8187
Deepfake-60K	Accuracy	0.9820	0.8496
	F1	0.9820	0.8461
	AUC	—	0.9922
	AP	—	0.9930

On Deepfake-vs-Real-60K, we get 84.96% accuracy with AUC 0.9922 and AP 0.9930. The high AUC and AP indicate near-perfect ranking, while the lower accuracy indicates threshold misplacement in prototype space. With only 50 support samples per class, prototype means may not represent tails of the class distribution, which can shift hard decisions even when embeddings are well separated.

Figure 3: Few-shot prototype adaptation metrics on AI vs Human Generated Images dataset.

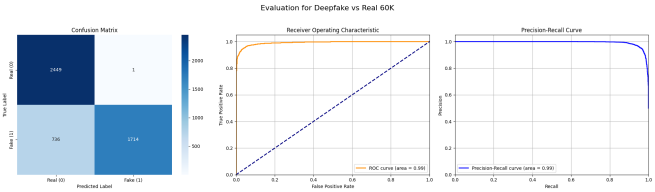

Figure 4: Few-shot prototype adaptation metrics on Deepfake vs Real 60K dataset.

For AI vs Human Generated Images, accuracy is 77.16% and AUC is 0.8451. Performance drops because this dataset includes broader generator styles and non-face content, which differs from HiDF's face-centered training distribution. The contrast with Deepfake-vs-Real-60K is consistent with domain proximity: face-oriented targets remain easier for the transferred encoder.

5.5 Assignment 3 — Extended Experiments

5.5.1 CiFake 140K Experiment

Scaling CiFake from 10K to 140K improves every reported metric in Table 8. Accuracy rises from 91.93% to 94.40%, AUC from 0.9721 to 0.9896, and F1 from 0.9196 to 0.9447. The gain suggests that larger supervision partly offsets reduced backbone capacity when the task distribution is stable.

Table 8: CiFake scaling: 10K (Assignment 2) vs. 140K (Assignment 3).

Metric	10K	140K	Paper
Accuracy	91.93%	94.40%	>95%
AUC	0.9721	0.9896	0.986
F1	0.9196	0.9447	∼0.950
AP	0.9712	0.9899	∼0.972

Compared with the paper, our 140K run nearly matches their reported accuracy level and exceeds their listed AUC/AP references. This pattern indicates that data scale can recover much of the lost capacity from moving ViT-L to ViT-B on CiFake.

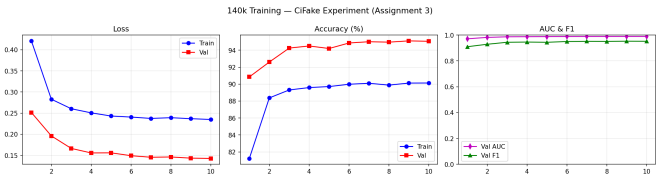

Figure 5: Training curves for CiFake classifier trained on 140K images (Assignment 3 extension).

Figure 5 shows stable convergence with validation saturation around epochs 7–8. Loss trends remain smooth, which suggests that regularization is adequate despite larger data volume.

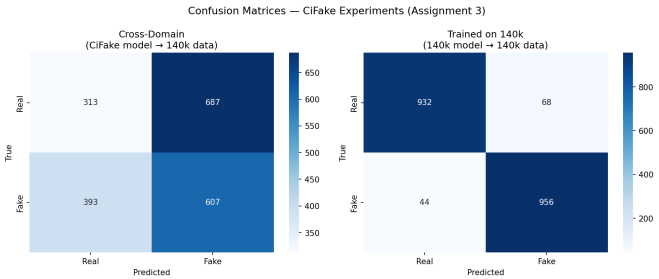

Figure 6: Confusion matrices for CiFake 140K experiment.

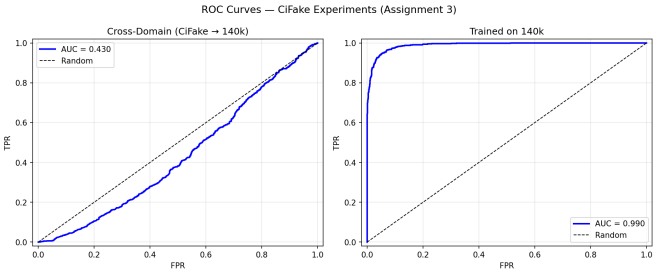

Figure 7: ROC curves for CiFake 140K experiment showing cross-domain evaluation performance.

Cross-domain evaluation of this same model on 140K Real & Fake Faces gives 46.0% accuracy, 0.4303 AUC, and 0.5292 F1. Because all three values are near or below random behavior, the learned head appears tied to CiFake-specific generator cues. The failure likely stems from stylistic mismatch between CiFake artifacts and StyleGAN-dominated facial synthesis.

5.5.2 HiDF Cross-Domain Evaluation

Table 9 reports HiDF-to-140K transfer. Accuracy is 53.90%, only 3.90 points above chance, which indicates weak out-of-domain discriminative power.

Table 9: HiDF cross-domain evaluation on 140K Real & Fake Faces.

Metric	Value
Accuracy	53.90%
Precision	0.5289
Recall	0.7130
F1	0.6073
CE Loss	0.9823
Total Images	2,000

Precision is 0.5289 while recall is 0.7130, so the model over-predicts the fake class on this target domain. The confusion pattern matches that bias, with many real images flagged as fake. One explanation is feature specialization to manipulation traces common in HiDF but not in StyleGAN synthesis.

5.5.3 SID_Set Experiments

We compare two SID-based runs in Table 10: in-domain binary training and cross-domain transfer to 140K Real & Fake Faces. Experiment 1 reaches 85.94% accuracy, while Experiment 2 drops to 50.05%. The 35.89-point drop indicates near-complete loss of usable domain transfer under this adaptation route.

Table 10: SID_Set Experiment 1 (in-domain) vs. Experiment 2 (cross-domain to 140K).

Metric	Exp 1 (In-domain)	Exp 2 (Cross-domain)
Accuracy	85.94%	50.05%
F1	0.8645	0.6667
AUC	—	0.5628
AP	—	0.5231

In the cross-domain run, the model assigns most samples to fake-related internal states, then maps them to a high fake probability. That behavior yields chance-level accuracy despite confident outputs, which indicates calibration bias rather than meaningful transfer.

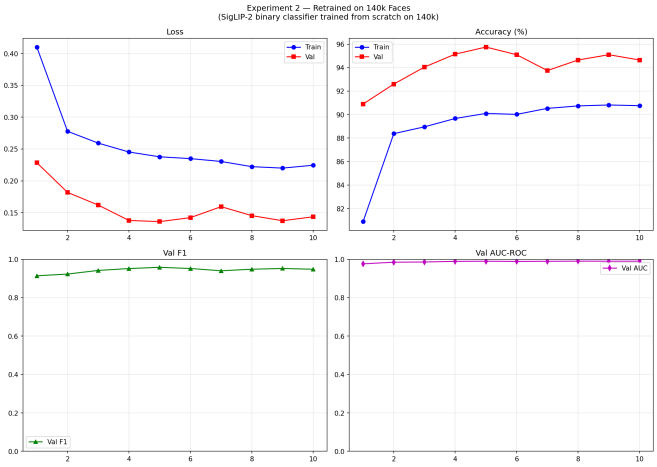

Figure 8: Training curves for SID_Set Experiment 2.

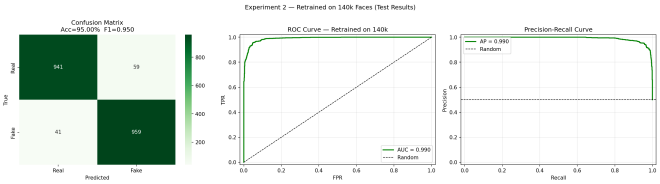

Figure 9: Test results for SID_Set Experiment 2.

Figures 8 and 9 indicate stable optimization but weak target-domain utility. Fast convergence with low transfer quality is consistent with overfitting to source-domain label semantics.

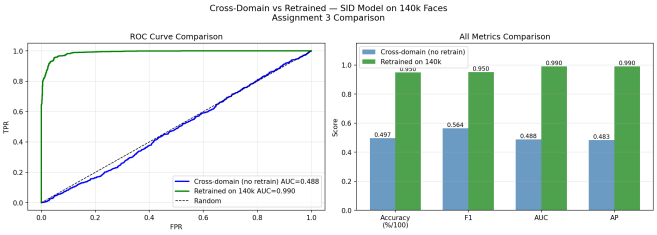

Figure 10: Side-by-side comparison of Experiment 1 and Experiment 2 results on SID_Set.

Figure 10 visualizes the in-domain versus cross-domain gap. The separation margin in Experiment 1 does not carry over to the target data, which suggests domain-specific decision geometry.

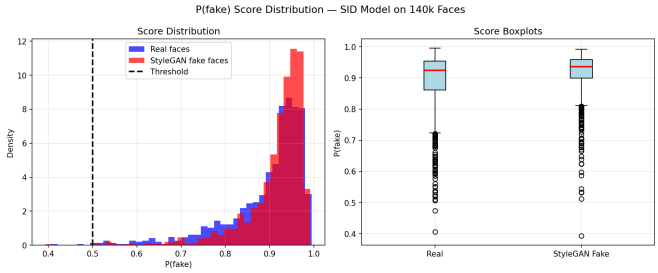

Figure 11: Prediction score distribution for SID_Set cross-domain evaluation.

Prediction scores in Figure 11 cluster near 0 and 1. This high confidence with low accuracy is a known out-of-distribution failure mode for neural classifiers.

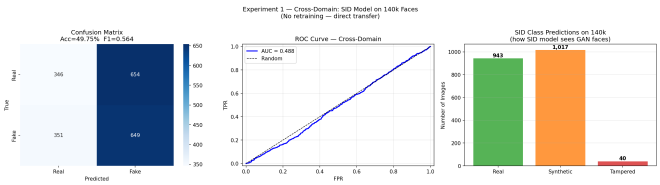

Figure 12: Cross-domain evaluation results for SID_Set Experiment 1.

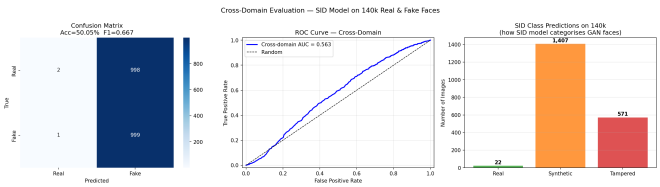

Figure 13: Cross-domain generalization results for SID_Set binary classifier.

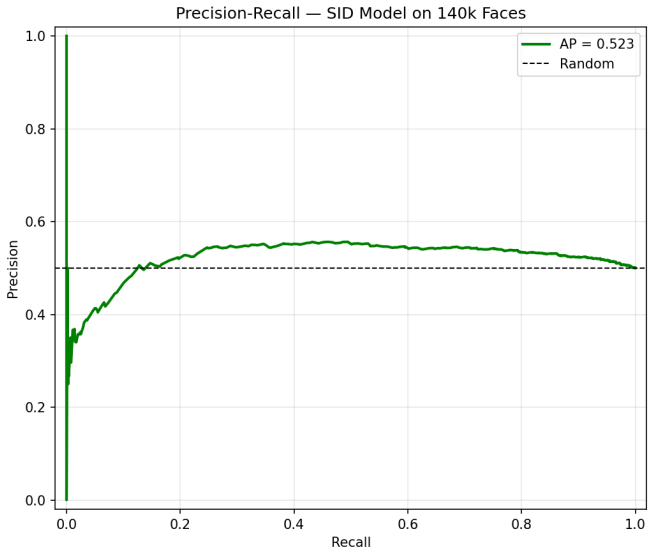

Figure 14: Precision-recall curve for SID_Set cross-domain evaluation.

Figures 12, 13, and 14 provide consistent evidence of transfer failure. AP is 0.5231, which stays close to random ranking and indicates little domain-invariant signal for StyleGAN-like samples.

6 Discussion

Our most important finding is that SigLIP-2 features remain useful under strict hardware limits, but transfer quality depends strongly on domain match. In-domain results stay competitive on CiFake and acceptable on HiDF and SID binary settings. Cross-domain behavior, however, drops to chance across all three trained heads. This contrast indicates that pretrained features alone do not guarantee robust deployment without adaptation.

Backbone scale matters most when the dataset contains broader manipulation variability. On HiDF, we trail the paper by 11.40 accuracy points (79.60% vs. ∼91%), which likely reflects both lower feature dimensionality and full freezing of ViT-B. On CiFake, the gap is smaller at 3.07 points in the 10K setup and narrows after scaling data. This pattern suggests that simpler binary distributions can compensate for reduced capacity through larger supervision.

Few-shot transfer reveals a second nuance. We obtain 0.9922 AUC on Deepfake-vs-Real-60K but only 84.96% accuracy, so representation quality and decision calibration diverge in this regime. One explanation is prototype estimation error with only 50 supports per class. The decision boundary then shifts even when class ranking stays strong.

Cross-domain tests expose the sharpest limitation. CiFake-to-140K reaches 46.0% accuracy, HiDF-to-140K reaches 53.90%, and SID-to-140K reaches 50.05%. Because all three are near chance, each trained head ap-

pears to encode source-specific artifacts rather than stable forgery principles. The difference is partly attributable to generator mismatch, especially between source manipulations and StyleGAN synthesis patterns.

A practical implication follows from this failure profile. Single-domain detectors should not be treated as universal filters in live settings. Few-shot adaptation helps when target content remains structurally similar to source data, as our Deepfake-vs-Real-60K results indicate. For broader shifts, adaptation likely needs stronger calibration control and explicit domain-diversity training.

Several limitations constrain our conclusions. We do not run ViT-L/16 directly, so we cannot fully separate architecture effects from data and compute effects. We also omit SID segmentation, which limits comparability to the original multi-task benchmark. Our few-shot evaluation uses 4,900-image subsets, and full-dataset runs may adjust absolute values. If we extend this work, we would prioritize larger support-set studies, calibration-aware prototype methods, and mixed-domain training that includes multiple generator families from the start.

7 Conclusion

Our reproduction confirms the central claim of Soundararajan and Xu [1]: SigLIP-2 features provide useful deepfake evidence even when training resources are limited. With ViT-B on a T4 GPU, we reach 91.93% on CiFake (10K), 79.60% on HiDF, and 85.94% on SID_Set binary classification. These values remain below the paper's ViT-L results but preserve the same overall trend across benchmarks.

Assignment 3 extensions add two concrete findings. Scaling CiFake to 140K improves performance to 94.40% accuracy and 0.9896 AUC, which indicates that data volume can recover part of the backbone gap. Cross-domain tests still collapse to chance-level behavior, which means current heads remain tightly coupled to source distributions.

A specific next step is to combine few-shot prototype adaptation with calibrated threshold tuning on small labeled target sets. That direction targets the exact failure mode we observe: strong ranking under shift with weak decision calibration.

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